INTERNATIONAL ECONOMIC REVIEW
Vol. 39, No. 3, August 1998

CYCLICAL EFFECTS OF GOVERNMENT'S EMPLOYMENT AND GOODS PURCHASES*

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Distinguishing between the goods purchase and employee compensation components of government spending is important. Shocks to government goods purchases and shocks to government employment have the opposite effects on private output, private employment and private investment. Moreover, with this distinction in place, quantitative analysis reveals that government spending is no longer a significant driving source of the U.S. business cycle. On the contrary, the components of government spending are procyclical in the U.S. data primarily because of their positive responses to the output fluctuations generated by technology shocks.

1. INTRODUCTION

The idea that government spending is an important stimulus to cyclical economic activity is a popular one. It is supported by a long history of economic analyses. Traditional Keynesian IS–LM analysis explains that an increase in government spending directly boosts aggregate demand and, given underemployment, leads to accommodating expansions in employment and output. Indeed, due to a positive marginal propensity to consume, consumption rises along with output, reinforcing the increase in aggregate demand, so that the rise in government spending has multiplier effects on employment and output. More recently, real business cycle (RBC) studies of government spending (e.g., Aiyagari et al., 1992, Christiano and Eichenbaum 1992, and McGrattan 1994a, 1994b) reaffirm its importance as an impulse to the business cycle, envisaging a transmission mechanism that is different from the Keynesian one. In the RBC model, an increase in government spending works through a negative effect on private wealth to invoke expansions in employment and output. Moreover, for sufficiently persistent rises in government spending, employment and output increase enough to permit the accumulation of capital, which in turn calls forth further expansions of employment and output in a manner reminiscent of Keynesian multiplier effects. Because shocks to government spending sharply differ from technology shocks by inducing opposite co-movements between labor and its average productivity, the RBC studies argue that government spending is especially important for explaining labor market dynamics over the cycle.

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[1] I thank Peter Ireland and two anonymous referees for their constructive suggestions on the research for this paper. Any remaining errors are my own.

The present study revisits the question of the effects of government spending on cyclical economic activity. It does so within the context of a quantitative RBC model of the U.S. economy. The aim of the study is to advance our understanding of government spending's effects on the business cycle by drawing a distinction between two of its components: purchases of final goods (and services) and compensation of employees.[2] Neither traditional Keynesian nor existing RBC studies draw distinctions like this one. Instead, for those studies government spending consists of final goods purchases only. A priori there are two reasons for why separating employee compensation from goods purchases could matter for business cycle analysis. First, the employee compensation component accounts for a significant share of government spending. In the U.S. economy (1950 : 1–1993 : 4), the average government spending share of employee compensation is 59 per cent while that of goods purchases is 41 per cent. Second, in theory, shocks to government purchases and government employment have dramatically different effects on the economy. An increase in government purchases, as existing RBC theories explain, is an *expansionary* impulse to private employment, output, and investment. On the contrary, a rise in government employment is a *contractionary* shock to private employment, output, and investment. These differences, to be explained more fully later, are due to the sectoral employment reallocations occurring with shocks to government employment but not with shocks to government purchases.

Surprisingly, the quantitative findings show that government spending is not in fact a significant driving force in U.S. cyclical fluctuations. These findings are surprising, not just because a priori each of the government spending's components impact so crucially on employment, but also because they starkly contrast with traditional Keynesian and previous RBC analyses. The contrast in findings across the present and previous RBC studies is traced to separating the employment compensation and goods purchase components of government spending. Treating government's spending on goods and employment as one homogeneous expenditure on goods only, as do the earlier RBC studies, leads to a sizeable overestimate of government's correct share in private output in the U.S. economy and, thus, of the true ability of government spending to generate U.S. cycles.

The model economy of the present study also differentiates between government's purchases of consumption and investment goods. This differentiation allows exploration of the potential importance of transmission channels for government goods purchases over and above the wealth channel mentioned above: utility effects of government's consumption purchases and productivity effects of government's capital. Furthermore, the model can explain the circumstances under which government investment crowds out private investment. The quantitative results reveal that increases in both the substitutability between government and private consumption

[2] Wynne (1992) and Rotemberg and Woodford (1992) are earlier studies making this distinction. Wynne (1992), the first to make the distinction, undertakes a comparative static analysis of government purchases and employment in a static deterministic neoclassical model. Rotemberg and Woodford (1992) compare oligopolistic and perfectly competitive models in terms of their impulse responses to military purchase and employment shocks.

and the productivity of government capital generally mitigate the ability of government consumption and government investment, respectively, to explain features of the U.S. data.

The quantitative theoretical investigation undertaken here necessarily involves estimation of the stochastic processes followed by four exogenous variables: technology, government consumption, government investment, and government employment. This exercise uses data on the U.S. economy (1950:1–1993:4). Estimates of the stochastic processes indicate that technology movements are not caused by changes in any of the three government variables. On the other hand, evidence of causality running from technology to each of the three government variables, especially to the investment and employment ones, is found, suggesting that government's decisions respond to the state of the economy as reflected in technology. This unidirectional pattern of causality has an interesting and important implication, identified through quantitative model analysis. The positive responses of the government variables to the state of the economy, as captured by technology, go far to explain why they, particularly government investment and government employment, are procyclical over the U.S. cycle. That is, the procyclicality of the government variables is due not so much to their own positive effects on output but rather to their positive responses to the output fluctuations generated by technology movements.

The remainder of the paper is organized as follows. Section 2 outlines the model economy. Section 3 provides a qualitative discussion of how government affects the model economy. Section 4 describes the model calibration and simulation. Section 5 presents the quantitative analysis, and Section 6 concludes.

2. THE MODEL ECONOMY

Consider an economy with a representative firm and household and a government. These agents interact in a variety of ways within a perfectly competitive market structure. Stochastic exogenous shocks to technology, government consumption, government investment, and government employment are the sources of fluctuations in the economy. A more exact description of the economy's structure and competitive equilibrium follows.

2.1. *Structure.* Taking market prices as given, the firm maximizes profit from goods production by optimally choosing labor and capital, that is:

$$\max_{(n_t^p,\, k_t^p)} \pi_t = y_t - w_t n_t^p - r_t k_t^p \tag{1}$$

subject to the production function:

$$y_t = F(z_t n_t^p, k_t^p, k_t^g) = (z_t n_t^p)^{\theta} (k_t^p)^{(1-\theta)} (k_t^g)^{\upsilon} \tag{2}$$

$$0 < \theta < 1, \qquad \upsilon > 0$$

The notation in equation (1) is: π is profit, y is output, w is the wage rate of labor, n^p is labor hired by the private firm, r is the rental rate of capital, and k^p is capital rented by the private firm and in place at the start of the period. In equation (2), z is an exogenous shock to technology, k^g is the government's capital, and θ and υ are parameters. The production function F displays the usual properties and constant returns to scale with respect to n^p and k^p. It differs from the standard neoclassical production function solely by the inclusion of k^g. Freely made available by the government at the beginning of the period, k^g is taken as given by the firm. The manner in which k^g enters into (2) follows Aschauer (1989), admitting a direct relationship between k^g and the productivity of n^p and k^p. This productivity effect of k^g is one channel through which the government influences the economy.

The infinitely lived household has preferences over consumption and labor that are defined in:

$$(3) \qquad E_0 \sum_{t=0}^{\infty} \beta^t U(c_t + \omega g_t, n_t)$$

$$U(\cdot) = \frac{\left[(c_t + \omega g_t)^{\psi}(1-n_t)^{(1-\psi)}\right]^{(1-\alpha)} - 1}{(1-\alpha)}$$

$$0 < \beta,\ \psi < 1, \qquad 0 \leq \omega \leq 1, \qquad \alpha > 0$$

where c and n are the household's consumption and labor supply, respectively, g is government consumption, β is the subjective discount factor, and ω, ψ, and α are other preference parameters. Available time each period is normalized at unity. The momentary utility function U exhibits standard properties, though it differs from the usual neoclassical utility function by including g. Aschauer and Greenwood (1985) similarly allow U to depend on g. The idea is that g provides utility in so far as it substitutes for c. The degree of substitutability across g and c is directly determined by ω. Thus, government can possibly affect the economy through the utility effect of g.

Each period, the household invests some of the economy's good to form the capital stock k^p in accordance with the law of motion:

$$(4) \qquad k_{t+1}^p = (1-\delta^p)k_t^p + i_t, \qquad 0 < \delta^p < 1$$

where i is the household's gross investment, and δ^p is a constant depreciation rate. Also each period, the household faces a budget constraint that sets its total income equal to its total spending:

$$(5) \qquad (1-\tau^l)w_t n_t + \left[r_t - \tau^c(r_t - \delta^p)\right]k_t^p + f_t + (1+q_t)b_t = c_t + i_t + b_{t+1}$$

$$0 < \tau^l, \quad \tau^c < 1$$

where f is a transfer payment from the government to the household, q is the net real interest rate, b is the household's stock of one-period bonds, each of which pays out $(1+q)$ units of goods upon maturity, and τ^l and τ^c are government-imposed tax rates on the household's labor and capital income, respectively. In equation (5), income comprises of after-tax labor and capital income, transfers and bond returns; spending is on consumption, investment, and new bonds. Capital income is not fully taxed—allowance is made for depreciation. The proportional tax rates on household income constitute additional avenues through which the government has economic effects.[3] The household's problem may now be stated. It is to maximize lifetime utility in (3) by optimally choosing c, n, and i subject to the constraints in (4)–(5). In doing so, the household takes market prices, g, and f as given.

Government's economic activities include purchasing goods for consumption and investment purposes, hiring labor, taxing income, and making transfer payments. Like the household, the government faces a budget constraint setting its total income equal to its total spending each period:

$$\tau^l w_t n_t + \tau^c (r_t - \delta^p) k_t^p + d_{t+1} = g_t + x_t + w_t n_t^g + f_t + (1+q_t)\, d_t \tag{6}$$

where d is the government's stock of one-period debt, each unit of which pays out $(1+q)$ units of goods upon maturity, x is government investment, and n^g is labor hired by the government. Equation (6) shows that income consists of income-tax revenue and the proceeds from new debt issue; spending is on consumption, investment, labor employment, transfers, and debt redemption. Government's investment creates the capital stock k^g subject to the law of motion:

$$k_{t+1}^g = (1-\delta^g) k_t^g + x_t, \qquad 0 < \delta^g < 1 \tag{7}$$

where δ^g is a constant depreciation rate. Exogenously to the model economy, the government chooses g, x, and n^g each period, and sets τ^l and τ^c once-and-for-all. Market prices, n, and k^p are taken as given by the government. Thus, from (6), f and/or d endogenously adjust to ensure that the budget constraint is satisfied.[4]

The last channel through which government influences the economy is now evident, namely, n^g. It is also worth emphasizing that even if g did not have utility effects and x, working through k^g, did not have production effects, both g and x would still impact on the economy. The reason is that g and x constitute absorptions of the economy's resources. In fact, when g does not substitute in utility for c (i.e., when $\omega = 0$), g has its maximum private wealth effect and, thus, its largest effect on the economy.

The specification of the stochastic processes of the four exogenous variables: z, g, x, and n^g, is as follows.

$$\log(S_t) = (I - A_1)\log(\bar{S}) + A_1 \log(S_{t-1}) + V_t \tag{8}$$

[3] Transfer payments have no effects on the economy. See footnote 4 for an explanation.

[4] As will be seen in the next subsection describing the competitive equilibrium, the time paths of f and d do not affect equilibrium allocations. This 'Ricardian Equivalence' feature implies that the government does not affect the economy through f or d.

where $S = [zgxn^g]'$, V is a (4×1) vector of innovations, I is a (4×4) identity matrix, A_1 is a (4×4) matrix of coefficients, and $\bar{S}$ is the mean of S. The innovation vector V is generated from a stationary, mean-zero, white-noise, normal distribution function. All of the diagonal elements of A_1 are positive fractions. Thus, S is a stationary, positively autocorrelated vector stochastic process.

The model economy imposes one identification assumption on the structure in (8): z_t is unaffected by the movements in g_{t-1}, x_{t-1} and n^g_{t-1}. That is, the only nonzero element in the first row of A_1 is the first one. Therefore, the model identifies the pure technological developments that arise independently of recent government decisions. No other restriction is imposed on (8). Thus, the model allows the three exogenous government variables not only to interact together but also to respond either contemporaneously or with a lag to the state of the economy as represented by technology. The vector stochastic process S is estimated and the model's identification restriction is tested using U.S. data (1950:1–1993:4). Section 4 reports the findings of the empirical exercise.

2.2. *Competitive Equilibrium.* Competitive equilibrium obtains when the firm and household solve their optimization problems, the government satisfies its budget constraint, and all markets clear. The equilibrium is implicitly determined by the laws of motion for k^p and k^g ((4) and (7)), the government budget constraint (6), the stochastic exogenous processes (8), and the following equations.

$$w_t = F_{n^p}(z_t n^p_t, k^p_t, k^g_t) \tag{9}$$

$$r_t = F_{k^p}(z_t n^p_t, k^p_t, k^g_t) \tag{10}$$

$$-U_n(c_t + \omega g_t, n_t) = U_c(c_t + \omega g_t, n_t)(1 - \tau^l) w_t \tag{11}$$

$$\begin{aligned} U_c(c_t + \omega g_t, n_t) & \\ &= \beta E_t\left[U_c(c_{t+1} + \omega g_{t+1}, n_{t+1})\{(1 - \tau^c) r_{t+1} + \tau^c \delta^p + (1 - \delta^p)\}\right] \end{aligned} \tag{12}$$

$$n_t = n^p_t + n^g_t \tag{13}$$

$$y_t = F(z_t n^p_t, k^p_t, k^g_t) = c_t + i_t + g_t + x_t \tag{14}$$

Equations (9) and (10), the outcomes of firm maximizing behavior, show that factor prices equal their respective marginal productivities. The household's intratemporal and intertemporal efficiency conditions governing its labor supply and investment are equations (11) and (12), respectively. Equation (13) states that the labor market clears—labor supply equals the sum of the firm's and government's labor demands. The economy's resource constraint is equation (14), showing that the consumption and investment of households and government completely absorb output. It derives from substituting the government budget constraint (6) into the household budget constraint (5), while noting the bond market clearing condition ($b_t = d_t$), labor market equilibrium (equation (13)), and the factor pricing equations ((9) and (10)).

3. QUALITATIVE WORKINGS OF GOVERNMENT SHOCKS

To provide intuition on how the government impacts on the model economy, particularly on private employment and its average productivity, this section discusses the key qualitative effects of one-time innovations to each of the three exogenous variables: g, n^g, and x. The discussion abstracts from interactions between these variables so as to clearly isolate their individual effects.

3.1. *Innovation to g*. Suppose a positive innovation increases g. So long as g is not perfectly substitutable for c in the utility of households, that is, $\omega < 1$, then the rise in g causes a negative private wealth effect. The reason is that the higher g means that more of the economy's goods are consumed by the government and, when $\omega < 1$, the marginal utility of g is less than the marginal utility of c.[5] Indeed, the marginal utility of g increasingly falls short of the marginal utility of c, the less well g substitutes for c or the lower is ω. Thus, the absolute size of the negative wealth effect is inversely related to the degree of substitutability between g and c, as measured by ω; being the largest when $\omega = 0$. This negative wealth effect is important—it is the main channel through which g exerts effects on the economy.

In response to the negative wealth effect, c falls and n rises; n rises through an increase in n^p. With k^p and k^g predetermined and z fixed, labor's average productivity (AP) declines as n^p rises. Therefore, AP and n^p move in opposite directions to one another. The increase in n^p causes y to expand. Whether i rises or falls crucially depends on the persistency of the increase in g. The more persistent is the rise in g, the more likely it is that i rises too. The reason is that the absolute value of the negative wealth effect is directly proportional to the degree of persistency in the expansion of g. Therefore, the more persistent is the increase in g, the greater is the fall in c and rise in n^p, both of which enhance the feasibility of an increase in i. Moreover, the expansion of n^p is more persistent, raising the incentive to invest by raising capital's future marginal productivity. Note that since g is positively autocorrelated, all of the effects discussed above (relative to the steady state) persist for some time. The following schematic summarizes the effects of the rise in g.

Exogenous Shock	*Endogenous Responses*
$\uparrow g$	$\uparrow y \uparrow n^p \downarrow AP \downarrow c \uparrow i$

3.2. *Innovation to n^g*. Next consider what happens when a positive innovation raises n^g. The increase in n^g has a negative effect on private wealth, because it expands government's usage of private resources. This negative wealth effect engenders a fall in c and a rise in n, as usual. But now, the increase in n occurs *only* through the increase in n^g, and n^p *decreases*. The reason is that the shock to n^g directly impacts on n; it is accommodated partially by an expansion of n and partially by a contraction of n^p. That is, there is a sectoral reallocation involving a

[5] When g perfectly substitutes for c, i.e., $\omega = 1$, the marginal utilities of g and c are the same and the rise in g has no effect on private wealth.

shift of labor out of the private sector and into the government sector. Simultaneously with the decline in n^p a rise in AP occurs, since z is fixed and k^p and k^g are predetermined. Thus, there is negative covariation between n^p and AP. The decrease in n^p causes y to contract. The more persistent is the expansion of n^g, the more likely it is that i contracts too. Although the fall in c augments the possibility of a rise in i, the fall in n^p tends to dominate in shrinking that possibility. In addition, when the increase in n^g is persistent, so also is the decline of n^p, which lowers the desire to invest by reducing capital's future marginal productivity. All of the foregoing effects (relative to the steady state) exhibit some persistency, stemming from the positive serial correlation of n^g. A summary of the effects of a rise in n^g is presented in the following schematic.

Exogenous Shock	*Endogenous Responses*
$\uparrow n^g$	$\downarrow y \downarrow n^p \uparrow AP \downarrow c \downarrow i$

3.3. *Innovation to x.* How the economy responds to a positive innovation increasing x depends on the shock's size relative to the productivity of k^g, that is, υ. This dependence stems from the fact that the rise in x has two opposing effects on private wealth. One effect, proportional to the x shock, is contractionary; it arises because the increase in x, like the earlier increase in g, means that more of the economy's goods are absorbed by government. The second effect on wealth, positively related to υ, is expansionary; arising since the higher x transforms over time into more k^g and, thus, enhances the productivity of private factors. Therefore, what happens to wealth depends on the size of the shock relative to υ. And this wealth effect determines much of the economy's subsequent course.

Suppose υ is high relative to the shock size, so that the rise in x has a positive wealth effect. Because k^g is predetermined, the economy's responses in the first period are quite different from those in later periods. In the first period, the positive wealth effect works to increase c and reduce n^p. This fall in n^p generates a rise in AP, given that z is fixed and k^g and k^p are predetermined. Opposite movements in n^p and AP therefore occur. As n^p declines, so too must y. The response of i is primarily dictated by feasibility considerations: the contraction of y together with the expansions in c and x force i to fall. That is, the increase in x 'crowds out' i. A summary of these first period responses is in the schematic:

Exogenous Shock	*First Period Endogenous Responses*
$\uparrow x$	$\downarrow y \downarrow n^p \uparrow AP \uparrow c \downarrow i$

Over time the higher x builds into higher k^g. Once k^g begins rising, substantial improvements to productivity take place. By directly raising labor's marginal productivity, the increase in k^g prompts a powerful intratemporal substitution effect that reinforces the rise in c and outweighs the income effect on n^p, causing n^p to rise. The higher k^g directly increases AP also, so that n^p and AP now move together. y increases in response to the increases in n^p and k^g. The two key forces operating to raise i are: the improvement in k^p's future marginal productivity due to the higher k^g, and the enhanced feasibility of expanding i because y has increased. Note that the positive autocorrelation of x implies that k^g rises for a protracted period of time. Consequently, k^g's effects on the economy (relative to the steady state) display strong persistence. The economy's second period and later responses to the x shock are summarized in the following schematic.

Exogenous Shock	*Second Period and Later Endogenous Responses*
$\uparrow x$	$\uparrow y \uparrow n^p \uparrow AP \uparrow c \uparrow i$

When the size of the shock to x is high relative to υ, so that the x increase generates a negative wealth effect, the pattern of economic events is different to that described above. But, the difference is manifested only in the first period. Because of the negative wealth effect, the rise in x initially acts just like a rise in g. Therefore, the economy's first-period responses are identical to those discussed earlier in the case of g's expansion. Economic effects in the second and later periods are once again driven by the increase in k^g and, thus, are the same as those described earlier in this subsection.

4. CALIBRATION AND SIMULATION

Quantitative exploration of the model's cyclical implications requires two preliminary steps: calibration and simulation. This section describes each step.

4.1. *Calibration.* The calibration procedure advanced by Kydland and Prescott (1982) is adopted. In this procedure, values are assigned to the model's parameters and steady state variables. These values are based on information drawn from a variety of sources, including U.S. data. An appendix describes the published U.S. data. The model's time period is defined as one quarter and the calibration recognizes this definition.

Table 1 presents some parameter and steady state variable values. Steady state variables are denoted using the same notation as before except that time subscripts are omitted. The values for β, α, θ, and n are the same as those often used in quantitative studies (e.g., King et al., 1988, Kydland and Prescott 1991, and Greenwood et al., 1992). Finn (1993) provides the estimate of υ. The effective average U.S. tax rates on labor and capital income computed in Mendoza et al., (1994) are the values of τ^l and τ^c. ω is set equal to zero, to allow exploration of the maximum effects of g. The present study, using the U.S. data, gives the estimates of δ^p and δ^g (see the Appendix for details). Average shares of government consumption and investment in private output and the average ratio of government to private employment, for the U.S. economy, provide the values of g/y, x/y, and n^g/n^p, respectively. Given the aforementioned number settings, together with the normalization of y at unity, the model's steady state relationships imply the numerical solutions for ψ and all remaining steady state variables.[6]

[6] An analysis of the effects of some alternative value settings for the central government parameters, υ, ω, τ^l, and τ^c, is also undertaken below.

TABLE 1
PARAMETER AND STEADY STATE VARIABLE VALUES

Preferences	Steady State Variables
$\beta = 0.99$	$y = 1.0000$
$\omega = 0.00$	$c = 0.7303$
$\alpha = 2.00$	$i = 0.1756$
$\psi = 0.23$	$n^p = 0.1695$
	$k^p = 7.0223$
	$k^g = 2.2800$
Production	
$\theta = 0.70$	$z = 2.1195$
$\nu = 0.16$	$g = 0.0713 = g/y$
$\delta^p = 0.025$	$x = 0.0228 = x/y$
$\delta^g = 0.01$	$n^g = 0.0305$, $n^g/n^p = 0.18$
	$n = n^p + n^g = 0.20$
Tax Rates	
$\tau^l = 0.25$	
$\tau^c = 0.43$	

The parameters of the stochastic exogenous processes in (8) are estimated from the U.S. data. This estimation involved several tasks. First, due to the absence of published U.S. data on z, an empirical U.S. z series was derived by 'solving' the production function in equation (2), given the calibrated values of θ and υ and the published U.S. data on y, n^p, k^p, and k^g. Second, because all four U.S. series on z, g, x, and n^g exhibit trending behavior, they were detrended to obtain their stationary components that more closely correspond to the model's stationary z, g, x, and n^g.[7] Third, using the stationary U.S. data on the four exogenous variables, the system of equations in (8) is estimated. Seemingly-unrelated regression (SUR), the estimation method, provides efficient parameter estimates in the presence of contemporaneous correlations between the innovations.

Table 2 displays the results from the unrestricted estimation of (8). z_t does not significantly depend on g_{t-1}, x_{t-1} or n^g_{t-1} individually. Moreover, the model's identification hypothesis that z_t is unaffected by g_{t-1}, x_{t-1} and n^g_{t-1} jointly cannot be rejected at the standard 10 per cent significance level.[8] Thus, the model's identification assumption is imposed on (8), permitting further gains in estimation efficiency. The findings from the restricted estimation of (8) are shown in Table 3. Notice that each exogenous variable is highly and positively autocorrelated. z_{t-1} has a positive effect on the three government variables, especially on x_t. All three government variables interact with one another, the most powerful connections

[7] Detrending was accomplished as follows. First, the logarithm of each exogenous variable was regressed on a constant and a polynomial-of-order-three in time. Second, the estimated polynomial time trend was subtracted from the logarithm of the respective exogenous variable to obtain its stationary component. A polynomial of up to order three, allowing for slowly evolving movements in trend, was necessary to extract the stationary components. Estimates of these trends are available upon request.

[8] The joint test, a likelihood ratio test, is distributed as a chi-squared statistic and has a marginal significance level equal to 0.49, which is bigger than the standard 0.10 threshold criterion.

TABLE 2
UNRESTRICTED SUR ESTIMATION RESULTS*

Eq.	Dep. Var.	Regressors	Coefficients	T-Statistics	Significance
(1)	z_t	z_{t-1}	0.90	25.50	0.00
		g_{t-1}	0.001	0.20	0.84
		x_{t-1}	0.003	0.29	0.78
		n^g_{t-1}	−0.03	−1.20	0.23
(2)	g_t	z_{t-1}	0.04	0.24	0.81
		g_{t-1}	0.86	30.05	0.00
		x_{t-1}	−0.07	−1.41	0.16
		n^g_{t-1}	0.39	3.11	0.002
(3)	x_t	z_{t-1}	0.28	2.44	0.01
		g_{t-1}	−0.01	−0.44	0.66
		x_{t-1}	0.88	25.70	0.00
		n^g_{t-1}	0.14	1.77	0.08
(4)	n^g_t	z_{t-1}	0.10	1.71	0.09
		g_{t-1}	−0.02	−1.80	0.07
		x_{t-1}	0.06	3.41	0.001
		n^g_{t-1}	0.83	20.16	0.00
		v^z_t	v^g_t	v^x_t	$v^{n^g}_t$
	v^z_t	$0.104E-3$	0.07	0.11	0.05
	v^g_t		$2.835E-3$	0.001	0.31
	v^x_t			$1.132E-3$	0.07
	$v^{n^g}_t$				$0.313E-3$

* Sample period of estimation: 1950:2–1993:4. Estimated constants for each equation are not reported. v^j is the innovation to variable j, for $j = z, g, x, n^g$. Correlations are above the diagonal, while variances are along the diagonal.

being the positive ones between g and n^g. The restricted estimates in Table 3 are used in calibrating the model economy.

4.2. *Simulation.* With the calibration complete, model simulation is possible. The steps of the simulation analysis follow. First, the model is log-linearized around its steady state and solved using standard methods for linear dynamic equations (see King et al., 1988). Second, given the model's solution, 1000 samples of 176 observations (the same size as the U.S. data sample) on each variable of interest are simulated. Each sample corresponds to a randomly generated sample of 176 observations on the innovation vector, V.[9] Third, the model's samples are filtered using the Hodrick–Prescott filter so as to distill the high-to-medium frequency movements linked to the business cycle.[10] Finally, second moments summarizing the model's cyclical behavior are computed from the filtered data and averaged across the 1000 samples so as to mitigate sampling error.

[9] Each sample initially had 276 observations, with the first observation using steady-state values of the state variables. To lessen dependency on initial conditions, the first 100 observations were discarded, leaving 176 observations.

[10] The smoothing parameter of the Hodrick-Prescott filter is set equal to 1600, the same number used in filtering the U.S. data (see Kydland and Prescott 1990).

TABLE 3
RESTRICTED SUR ESTIMATION RESULTS*

Eq.	Dep. Var.	Regressors	Coefficients	T-Statistics	Significance
(1)	z_t	z_{t-1}	0.89	26.15	0.00
(2)	g_t	z_{t-1}	0.04	0.23	0.81
		g_{t-1}	0.86	30.11	0.00
		x_{t-1}	−0.07	−1.43	0.15
		n^g_{t-1}	0.40	3.20	0.001
(3)	x_t	z_{t-1}	0.28	2.43	0.02
		g_{t-1}	−0.01	−0.46	0.64
		x_{t-1}	0.88	25.82	0.00
		n^g_{t-1}	0.15	1.90	0.06
(4)	n^g_t	z_{t-1}	0.10	1.70	0.09
		g_{t-1}	−0.02	−1.81	0.07
		x_{t-1}	0.06	3.40	0.001
		n^g_{t-1}	0.84	20.25	0.00
		v^z_t	v^g_t	v^x_t	$v^{n^g}_t$
	v^z_t	$0.105E-3$	0.07	0.11	0.05
	v^g_t		$2.836E-3$	0.001	0.31
	v^x_t			$1.133E-3$	0.07
	$v^{n^g}_t$				$0.313E-3$

* See notes to Table 2.

5. QUANTITATIVE ANALYSIS

Characteristics of U.S. cyclical fluctuations (1950:1–1993:4) are summarized in Table 4. To identify cyclical movements, the U.S. data (described in the Appendix) have been Hodrick–Prescott-filtered. Some features of the U.S. cycle are well known. Specifically, c and AP are less volatile, n^p is approximately as volatile, and i is more volatile than y. Each of c, i, n^p, and AP are strongly procyclical. And, the

TABLE 4
U.S. CYCLICAL CHARACTERISTICS*

	SD	*CY*
y	1.77	1.00
c	0.80	0.81
i	4.63	0.76
n^p	1.82	0.85
AP	0.97	0.23
z	1.54	0.70
g	10.23	0.04
x	4.43	0.08
n^g	2.59	0.16
	$c(n^p, AP) = -0.31$	

* Sample period of data is 1950:1–1993:4; SD denotes percentage standard deviation; CY is the correlation between the left-side variable and y; $c(.,.)$ denotes correlation between variables in parentheses.

covariation between n^p and AP is negative.[11] Less well known are the features of the government variables in the U.S. cycle. g, x, and n^g are substantially more volatile than y. All three government variables are procyclical, but just mildly so.

The questions of prime concern to the present study are: does government significantly influence the U.S. cycle and, if so, which of government's activities: g, x, and n^g provide the most important impulses? Answers to these questions are pursued through a quantitative analysis of the model economy outlined earlier. The quantitative analysis begins by examining the cyclical contribution of independent shocks to g, x, and n^g. The sensitivity of this contribution to the magnitudes of critical 'government parameters' is checked. Next, the importance of government's size, measured by its expenditure and labor shares, is brought to the forefront. The issues involved here lead to an illustration of why the distinction between government's goods purchases and employment matters substantively. Finally, the quantitative analysis assesses the full cyclical contribution of g, x, and n^g when account is taken of their rich patterns of interaction with one another and with the state of the economy as captured by z.

5.1. *Isolating the Contribution of Government Shocks.* This subsection isolates the business cycle effects of independent variations in the three government variables. To achieve this isolation, the model economy is simulated with a stochastic shock structure that simplifies the system in equation (8) along two dimensions: (i) any nonzero and nondiagonal element of A_1 is set equal to zero and, (ii) all of the nonzero contemporaneous correlations among the innovations are equated to zero. With these restrictions in place, the four exogenous variables: z, g, x, and n^g are independent of each other. In addition, the model economy is simulated under alternative assumptions about which, if any, of the three government variables are subject to stochastic shocks. That is, sometimes the model simulation holds some of the government variables constant. Table 5 presents the results of the simulation. Each panel in the table indicates which, if any, of the government shocks are operating by showing whether the standard deviation of the innovation to g, x, and n^g (denoted by σ_j, $j = g, x, n^g$) is positive or zero.

Comparison of panel A to the U.S. data in Table 4 reveals that the model economy driven exclusively by shocks to z mimics many of the popular features of the U.S. cycle. However, there are some glaring discrepancies too—all involving n^p. The model significantly understates the volatility of n^p and, due to its reliance on z shocks, well overstates both the correlations between AP and y and between AP and n^p. If shocks to the government variables play an important role in the U.S. cycle, then incorporating them into the model should promote the match between the model and the U.S. facts. Most especially, the government shocks potentially explain the behavior involving n^p. For as was discussed in Section 3, each of g, x, and n^g not only impact on n^p making it more volatile, but also induce opposite movements of AP and y as well as of AP and n^p.

[11] Christiano and Eichenbaum (1992) argue that measurement error biases the correlation between n^p and AP downward so that the 'true' correlation is approximately zero.

TABLE 5
ISOLATING THE CONTRIBUTION OF GOVERNMENT SHOCKS*

	A. $\sigma_g, \sigma_x, \sigma_{n^g} = 0$		B. $\sigma_g, \sigma_x, \sigma_{n^g} > 0$	
	SD	*CY*	*SD*	*CY*
y	1.47	1.00	1.49	1.00
c	0.52	0.95	0.54	0.89
i	6.38	0.99	6.67	0.92
n^p	0.82	0.99	0.87	0.97
AP	0.68	0.98	0.69	0.95
z	1.29	0.998	1.29	0.989
g			7.78	0.11
x			4.18	0.00
n^g			2.15	−0.07
	$c(n^p, AP) = 0.93$		$c(n^p, AP) = 0.83$	

	C. Only $\sigma_g > 0$		D. Only $\sigma_x > 0$		E. Only $\sigma_{n^g} > 0$	
	SD	*CY*	*SD*	*CY*	*SD*	*CY*
y	1.48	1.00	1.47	1.00	1.48	1.00
c	0.54	0.89	0.52	0.95	0.52	0.95
i	6.62	0.92	6.41	0.99	6.41	0.99
n^p	0.85	0.97	0.82	0.99	0.84	0.98
AP	0.68	0.96	0.68	0.98	0.68	0.97
z	1.29	0.992	1.29	0.998	1.29	0.995
g	7.78	0.11				
x			4.18	0.00		
n^g					2.15	−0.07
	$c(n^p, AP) = 0.86$		$c(n^p, AP) = 0.93$		$c(n^p, AP) = 0.90$	

* See Table 4 for notes. σ_j is the standard deviation of innovations to variable j, $j = g$, x and n^g.

By comparing panels C, D and E to panel A, the separate effects of the independent government shocks may be seen. The joint effects of all three independent g, x, and n^g shocks are evident from a comparison of panels A and B. Even when the independent shocks to the three government variables are combined, they do not substantively bring the model's cyclical behavior into closer alignment with the U.S. data's. The strongest contribution comes from g in terms of reducing the correlation between AP and n^p, but it is not nearly powerful enough to rationalize the negative U.S. AP-n^p correlation.

Recall from Section 3 that the model predicts that increases in g raise y, x's effects on y are ambiguous, and expansions of n^g contract y. These effects are illustrated in panel B—g is procyclical, x is acyclical, and n^g is countercyclical. The model economy with independent variations in z, g, x, and n^g well explains the procyclicality of g, but cannot capture the procyclicality of both x and n^g that is evident in the U.S. data. Overall, the findings of this subsection suggest that government does not significantly cause cyclical fluctuations in the U.S. economy.

5.2. *Sensitivity to Central Parameters.* Here the sensitivity of government's cyclical contribution to the values of certain key parameters is analyzed. The parameters are those gauging channels of government's influence on the economy: ω, υ, τ^l, and τ^c. Specifically, the model economy with independent exogenous shock processes is simulated for values of ω, υ, τ^l, and τ^c differing from their benchmark

TABLE 6
SENSITIVITY TO CENTRAL PARAMETERS*

	A. $\omega = 1$		B. $\upsilon = 0.32$	
	SD	*CY*	*SD*	*CY*
y	1.51	1.00	1.49	1.00
c	0.94	0.55	0.55	0.90
i	6.48	0.99	6.68	0.92
n^p	0.87	0.98	0.85	0.97
AP	0.67	0.97	0.70	0.95
z	1.29	0.995	1.31	0.989
g	7.78	0.00	7.70	0.10
x	4.18	0.00	4.21	−0.03
n^g	2.15	−0.07	2.06	−0.07
	$c(n^p, AP) = 0.91$		$c(n^p, AP) = 0.84$	
	C. Low Taxes		D. High Taxes	
	SD	*CY*	*SD*	*CY*
y	1.50	1.00	1.47	1.00
c	0.53	0.88	0.54	0.89
i	6.51	0.92	6.87	0.92
n^p	0.88	0.97	0.85	0.97
AP	0.68	0.94	0.69	0.95
z	1.29	0.989	1.29	0.988
g	7.78	0.11	7.78	0.11
x	4.18	0.00	4.18	0.00
n^g	2.15	−0.07	2.15	−0.07
	$c(n^p, AP) = 0.83$		$c(n^p, AP) = 0.83$	

* See Table 4 for notes. Low taxes: $\tau^l = 0.18$, $\tau^c = 0.37$. High taxes: $\tau^l = 0.29$, $\tau^c = 0.49$.

values (listed earlier in Table 1) as follows:

	New Values	*Benchmark Values*
	$\omega = 1$, $\upsilon = 0.32$	$\omega = 0$, $\upsilon = 0.16$
Low taxes:	$\tau^l = 0.18$, $\tau^c = 0.37$	$\tau^l = 0.25$, $\tau^c = 0.43$
High taxes:	$\tau^l = 0.29$, $\tau^c = 0.49$	

Changing ω from zero to one implies that the substitutability of g for c in utility changes from being nonexistent to perfect. The higher new value of υ, making government capital much more productive, is at the upper end of the two standard deviation interval estimated in Finn (1993). The low and high tax rate combinations, enclosing the benchmark one, include the lowest and highest values, respectively, of τ^l and τ^c computed by Mendoza et al., (1994) for the United States over the period 1965–1988. Findings of the simulations with the new parameter values are shown in Table 6. The table's panels indicate which parameter values have been changed relative to the benchmark case by displaying the new parameter values.[12]

[12] Also, for the simulation with $\upsilon = 0.32$, values of the parameters of the stochastic exogenous processes in equation (8) differ from their benchmark values in Table 3. The reason is that the empirical U.S. z series changes as the value of υ does. Thus, the estimation exercise described in Section 4 was repeated using the new U.S. z series corresponding to $\upsilon = 0.32$. The resultant new parameterization of the stochastic shock structure is similar to the old one. These estimation findings are available upon request.

To assess the effects of the alterations in parameter values, compare Table 6 to the benchmark economy of panel B in Table 5. The changes in tax rates do not materially influence the model's cyclical characteristics. Raising υ has the one main effect of making x countercyclical. This countercyclicality stems from the fact that υ's rise switches the wealth effect of x increases from zero to positive. And, as Section 3 explained, wealth effects then operate to cause declines in y whenever x rises. The most noticeable effects of setting ω equal to one are: g becomes acyclical, the correlation between AP and n^p increases, c is more volatile and covaries less with y. These effects are also due to a change in wealth effects. When g perfectly substitutes for c, g's effect on wealth, the main channel of its influence, disappears. Thus, when ω equals one, g does not impact much on the economy. Instead, c moves to offset most of the changes in g.

In summary, the sensitivity analysis shows that only the changes in ω and υ notably affect government's cyclical contribution. But, making g more substitutable for c (i.e., increasing ω) or government capital more productive (i.e., raising υ) worsens the correspondence between the model and U.S. data. Thus, the benchmark parameterization of the model economy allows more leeway for government shocks to explain U.S. cycles.

5.3. *Why Government's Shares Matter*. Because g, x, and n^g are so volatile, it is perhaps surprising that they do not figure more prominently in the U.S. cycle. However, volatility is only one dimension auguring for the importance of government's cyclical effects. Size, measured by government's expenditure and labor shares, is another dimension. And, these shares are sufficiently small so as to constrain the ability of g, x, and n^g to significantly drive fluctuations in the U.S. economy.

To see this point more clearly, this subsection simulates the model economy, with independent exogenous shock processes, under the counterfactual assumptions that government's shares are much larger than they truly are in the U.S. and benchmark economies. The assumed increase in share values is as follows:

New Values	*Benchmark Values*
$s_g = 0.14$	$s_g = 0.07$
$s_x = 0.10$	$s_x = 0.02$
$n^g/n^p = 0.30$	$n^g/n^p = 0.18$

where s_g and s_x are the shares of g and x, respectively, in y. Table 7 presents the simulation results, with each panel indicating which government share has been increased above its benchmark value by displaying the new share value.

A comparison of Table 7 to the benchmark economy of panel B in Table 5 shows the effects of raising government's shares. Of particular note are the effects of the larger shares in terms of: increasing the volatility of n^p (generally) and weakening the covariations between AP and y and between AP and n^p.[13] In addition, g is

[13] One exception is that the volatility of n^p declines as the share of x rises. This finding reveals that while x shocks do impact on n^p, they are not as important as the other shocks in the economy for the behavior of n^p.

TABLE 7
SHOWING HOW GOVERNMENT'S SHARES MATTER*

	A. $s_g = 0.14$		B. $s_x = 0.10$		C. $n^g/n^p = 0.30$	
	SD	*CY*	*SD*	*CY*	*SD*	*CY*
y	1.49	1.00	1.46	1.00	1.51	1.00
c	0.60	0.74	0.57	0.86	0.54	0.88
i	7.26	0.74	6.77	0.88	6.82	0.92
n^p	0.93	0.93	0.84	0.95	0.92	0.96
AP	0.71	0.88	0.70	0.93	0.68	0.93
z	1.29	0.97	1.29	0.986	1.29	0.98
g	7.78	0.22	7.78	0.11	7.78	0.11
x	4.18	0.00	4.18	0.07	4.18	0.00
n^g	2.15	−0.07	2.15	−0.07	2.15	−0.11
	$c(n^p, AP) = 0.64$		$c(n^p, AP) = 0.78$		$c(n^p, AP) = 0.78$	

*See Table 4 for notes. s_g and s_x, respectively, denote the shares of government consumption and investment in private output.

more procyclical, x becomes procyclical, and n^g is more countercyclical. The procyclicality of x is produced from the phenomenon that the rising share of x switches the wealth effect of x increases from zero to negative. As was explained in Section 3, wealth effects then work to induce expansions in y whenever x expands. The other mentioned effects of bigger government shares are understandable from the earlier discussions of how g, x, and n^g impact on the economy and by noting that any shock to g, x, or n^g now translates into a larger absolute change. A shock is measured as a *percentage deviation* from the variable's average value—so it is absolutely bigger when the variable's share is. Thus, government's shares are seen to matter substantially. Their values in the U.S. economy are small enough to prevent the highly volatile g, x, and n^g processes from significantly influencing U.S. cyclical fluctuations.

5.4. *Mistaking the Contribution of g Shocks*. Seeing how the higher share of g promotes the strength of g's cyclical effects above, provides a hint as to why existing RBC studies (e.g., Christiano and Eichenbaum 1992, and McGrattan 1994a, 1994b) argue that g is an important impulse to U.S. business cycle fluctuations and, particularly, to U.S. labor market dynamics. The reason is that those studies do not distinguish between government's spending on consumption goods and labor employment, and instead treat the sum of the two types of spending as a homogeneous expenditure on consumption goods only. One implication of this aggregation is that the RBC studies of g shocks overstate g's correct share in the U.S. economy and, consequently, the true contribution of g shocks to U.S. cycles.

For instance, consider a model economy identical to the one in Section 2 except that n^g is always zero. Add wage compensation of government employees to government consumption to obtain an expanded U.S. measure of g. Calibrate the new model economy to the same preset parameter and steady state variable values listed in Table 1 with the following exceptions. The new preset steady state values are: $n^g = 0$ and $s_g = g = 0.21$, which is the average share of the augmented g in y for the U.S. economy. And, the new parameter estimates of the independent

TABLE 8
MISTAKING THE CONTRIBUTION OF g SHOCKS*

	A. $\sigma_g = 0$		B. $\sigma_g > 0$	
	SD	*CY*	*SD*	*CY*
y	1.43	1.00	1.45	1.00
c	0.56	0.95	0.60	0.82
i	6.36	0.995	6.80	0.85
n^p	0.76	0.98	0.85	0.94
AP	0.70	0.98	0.72	0.91
z	1.30	0.999	1.30	0.98
g			3.91	0.18
	$c(n^p, AP) = 0.92$		$c(n^p, AP) = 0.72$	

* See Tables 4 and 5 for notes.

components of the z, g, and x processes are:

$$z_t = 0.89 z_{t-1} + v_t^z, \qquad \sigma_z^2 = 0.106E-3$$

$$g_t = 0.87 g_{t-1} + v_t^g, \qquad \sigma_g^2 = 0.538E-3$$

$$x_t = 0.93 x_{t-1} + v_t^x, \qquad \sigma_x^2 = 1.149E-3$$

where v^j is the innovation to variable j $(j = z, g, x)$ and constant terms have been omitted for simplicity. New estimates of the stochastic exogenous shock processes are necessary primarily because of the new measure of g in the U.S. data.[14]

The findings from simulating the new model economy are in Table 8. Specifically, in panel A the model is driven solely by shocks to z, with g and x held constant. Variations to g (only) are added to those of z in panel B. Thus, comparing panels A and B illustrates the effects of g shocks. g strongly enhances the volatility of n^p, moderates the linkage between AP and y, and significantly reduces the correlation between AP and n^p. The power of these effects of g sharply contrasts with g's weak effects earlier in this section. The contrast is especially remarkable because the volatility of g is now substantially lower than it was before (compare σ_g^2 across the two cases). But, to conclude like the aforementioned RBC studies that g is, therefore, an important driving force of U.S. fluctuations would be a mistake—one predicated from not distinguishing between government's spending on employment and consumption.

5.5. *Allowing Interactions Between Shock Processes.* To assess the full cyclical contribution of g, x, and n^g, this subsection simulates the original model economy taking account of the patterns of interaction between z, g, x, and n^g evident from the estimates of the stochastic shock structure, equation (8), presented in Table 3.

[14] The new estimates were derived by repeating the estimation exercise described in Section 4 and the method of extracting independent components described earlier in this section, for the three variable system consisting of z, g, and x. This estimation uses the sum of government consumption and wage compensation to measure g in the U.S. data. The full set of estimation results is available upon request.

TABLE 9
ALLOWING INTERACTIONS BETWEEN SHOCK PROCESSES*

	A. Independence		B. Full Interaction		C. Partial Interaction	
	SD	*CY*	*SD*	*CY*	*SD*	*CY*
y	1.49	1.00	1.54	1.00	1.48	1.00
c	0.54	0.89	0.49	0.87	0.54	0.88
i	6.67	0.92	7.11	0.91	6.71	0.91
n^p	0.87	0.97	0.94	0.98	0.85	0.97
AP	0.69	0.95	0.66	0.95	0.69	0.96
z	1.29	0.989	1.29	0.992	1.29	0.992
g	7.78	0.11	8.38	0.12	8.37	0.10
x	4.18	0.00	4.28	0.14	4.11	0.02
n^g	2.15	−0.07	2.28	0.08	2.22	0.02
	$c(n^p, AP) = 0.83$		$c(n^p, AP) = 0.87$		$c(n^p, AP) = 0.86$	

* See Table 4 for notes.

Thus, the nondiagonal elements of A_1 and the correlations between innovations are reinstated, with the joint effect of inducing strong positive correlations between all pairs of: z, g, x, and n^g.

The simulation results are displayed in Table 9. For comparison purposes, panel A reproduces panel B of Table 5—the case of independent exogenous processes. In panel B, full interaction between z, g, x, and n^g occurs. Also for comparison purposes, panel C shows what happens when all interactions except those involving z are allowed (i.e., the nondiagonal elements of A_1 and innovation correlations connecting z to g, x, and n^g are set equal to zero).

Comparing panels A and B shows that the full interaction slightly raises the volatility of n^p and the AP-n^p correlation. n^p is more variable because g, x, and n^g are. Although each of g, x, and n^g induce opposite comovement of AP and n^p, they have different effects on both AP and n^p. Therefore, the positive correlations among g, x, and n^g undo some of the power that each of these shocks separately have to cause negative covariation between AP and n^p, so the correlation between AP and n^p rises. The remaining most noticeable effects of admitting full interactions is that x and n^g become procyclical. This procyclicality is largely the outcome of the fact that x and n^g are positively correlated with z, increases in which raise y. Absent these positive correlations with z, both x and n^g are essentially acyclical, as may be seen from panel C. Despite the fact that increases in n^g reduce y, n^g is acyclical in panel C because of its positive correlation with g, a shock that directly affects y. Shutting off n^g's positive correlation with g, in panel A, results in n^g being countercyclical.

The foregoing results show that accounting for the interactions between the four exogenous variables does not do much to close the gap between the model's cyclical characteristics and the U.S. data's. The only exceptions to this rule are that the interactions permit the model economy to well explain the procyclicality of x and n^g evident in the U.S. data. But the model identifies that the procyclicality of x and n^g is due to their positive correlations with z and/or g rather than to their own direct effects on y. Thus, the initial assessment in subsection (5.1) that g, x, and n^g do not play a prominent role in driving U.S. business cycles remains intact when the interactions between z, g, x, and n^g are fully accounted for.

6. CONCLUSION

This study explores the effects of government spending on the business cycle within the framework of a quantitative real business cycle model of the U.S. economy. A new feature of this exploration is the distinction between two broad categories of government spending: final goods purchases and employee compensation. Shocks to government purchases and to government employment have opposite effects on private employment, output, and investment. Also differentiated are government's purchases of consumption and investment goods so as to, especially, allow the role of government investment in enhancing the productivity of private factors of production to be taken into account. The quantitative investigation reveals that government spending is not a prominent driving force of the U.S. cycle. Government's size relative to the economy's, measured by government's purchase shares in private output and the ratio of government to private employment, is sufficiently small so that shocks to the components of government spending are constrained in their ability to strongly induce cyclical fluctuations. To the contrary, the quantitative analysis finds that most components of government spending are procyclical in the U.S. data largely because of their positive responses to the output fluctuations sparked by movements in technology.

It must be emphasized that these findings pertain only to the cyclical effects of government spending. They carry no implication regarding government's effects on the overall *level* of economic activity. Government can and does substantially promote the level of economic activity in a variety of ways, for example, providing infrastructure, advancing education and research, and maintaining law and order. Instead, the findings here suggest that government spending is not, contrary to popular opinion, a prime instigator of the short-run periodic fluctuations in economic activity around its average trending level.

7. APPENDIX

This study's data are quarterly, real, seasonally-adjusted, per-capita data for the United States from 1950:1–1993:4. DRI is the source, unless otherwise indicated. The second data source is the Wealth Data available from the National Income and Wealth Division, BEA, Department of Commerce (here referred to as DC). A description of the data follows.

Population (millions of persons). The sum of the noninstitutional aged-sixteen-and-over population (including the resident armed forces) and the overseas armed forces.

Aggregate Price Deflator (1987 = 1). Gross domestic product price deflator.

Private Output (billions of 1987 dollars). Gross domestic product less the component due to general government. In the national income accounts, general government is government exclusive of government enterprises.

Private Consumption (billions of 1987 dollars). Personal consumption expenditures on nondurable goods and services.

Private Investment (billions of 1987 dollars). Gross domestic fixed nonresidential investment by the private sector and government enterprises plus the same type of investment that is government owned but privately operated (here referred to as GOPO investment). Enterprise and GOPO investment are available only annually from DC. Quarterly enterprise and GOPO investment series were obtained by interpolating the annual ones. The interpolation procedure is described at the end of this Appendix.

Government Consumption (billions of 1987 dollars). Government purchases of goods and services less the sum of components attributable to enterprise and GOPO investment (described above), government investment (described below), the Commodity Credit Corporation inventory change, and the wage compensation of general government employees. This real series was obtained by deflating the component nominal series by the aggregate price deflator.

Government Investment (billions of 1987 dollars). Gross domestic fixed nonresidential nonmilitary investment by government less the component comprising of enterprise and GOPO investment. Like enterprise and GOPO investment, government investment is available only on an annual basis from DC. A quarterly government investment series was derived from the annual one through interpolation. The method is outlined below.

Private Labor Hours (millions of hours). The sum of labor hours in the private sector (Establishment Survey) and government enterprises. Enterprise labor hours are the product of enterprise employment (National Income and Product Accounts) and hours per government worker per quarter (Establishment Survey). Enterprise employment is published only at an annual frequency. A quarterly enterprise employment series was derived by linear interpolation of the annual one. Hours per government worker per quarter is an average measure across all government civilian workers, including those in the enterprises.

Government Labor Hours (millions of hours). Government civilian labor hours (Establishment Survey) less the component consisting of enterprise labor hours (described above) plus military hours. The series for military hours is the product of military employment (Population Survey) and hours per government worker per quarter (described above). Military employment was obtained by subtracting the civilian population from the total population inclusive of the armed forces overseas.

Private Capital Stock (billions of 1987 dollars). Net domestic fixed nonresidential capital stock of the private sector and government enterprises plus the same type of capital that is government owned but privately operated (GOPO capital). The components of this capital stock are available only on an annual basis from DC. A quarterly private capital series was derived in the following fashion. First, the quarterly depreciation rate of private capital was estimated. This quarterly estimate is one quarter of the annual depreciation rate estimated from the private capital accumulation equation, i.e. (4) in the text, using annual time series on private investment and capital (described above). Second, the quarterly private capital stock series was obtained by iterating on equation (4), using the quarterly private investment series, the quarterly depreciation rate and the value of the private capital stock at the end of 1949.

Government Capital Stock (billions of 1987 dollars). Net domestic fixed nonresidential nonmilitary capital stock of government less the component consisting of enterprise and GOPO capital. As in the case of private capital, the components of government capital are available only annually from DC. A quarterly government capital series was obtained in a manner analogous to that of the quarterly private capital series. In doing so, annual time series on government capital (described above) and both annual and quarterly time series on government investment (described above) were used.

Interpolation. To derive quarterly government, enterprise and GOPO investment series from the corresponding annual series, the interpolation procedure uses information in government total purchases. This information is useful because the government total purchase series is available at a quarterly frequency and it includes the three government-related investment series. Specifically, the interpolation rule equates the fraction of annual government, enterprise or GOPO investment occurring in any given quarter to the proportion of annual government purchases taking place in that same quarter.

REFERENCES

AIYAGARI, S. R., L. J. CHRISTIANO, AND M. EICHENBAUM, "The Output, Employment, and Interest Rate Effects of Government Consumption," *Journal of Monetary Economics* 30 (1992), 73–86.

ASCHAUER, D. A., "Is Public Expenditure Productive?" *Journal of Monetary Economics* 23 (1989), 177–200.

——— AND J. GREENWOOD, "Macroeconomic Effects of Fiscal Policy," *Carnegie-Rochester Conference Series on Public Policy* 23 (1985), 91–138.

CHRISTIANO, L. J. AND M. EICHENBAUM, "Current Real-Business-Cycle-Theories and Aggregate Labor Market Fluctuations," *American Economic Review* 82 (1992), 430–450.

FINN, M. G., "Is All Government Capital Productive?" *Economic Quarterly* Federal Reserve Bank of Richmond 79 (1993), 53–80.

GREENWOOD, J., Z. HERCOWITZ, AND P. KRUSELL, "Macroeconomic Implications of Investment-Specific Technological Change," Discussion Paper 76, Institute for Empirical Macroeconomics, Federal Reserve Bank of Minneapolis, 1992.

KING, R. G., C. I. PLOSSER, AND S. T. REBELO, "Production, Growth and Business Cycles," *Journal of Monetary Economics* 21 (1988), 195–232.

KYDLAND, F. E. AND E. C. PRESCOTT, "Time to Build and Aggregate Fluctuations," *Econometrica* 50 (1982), 1345–1370.

——— AND ———, "Business Cycles: Real Facts and a Monetary Myth," *Quarterly Review* Federal Reserve Bank of Minneapolis 14 (1990), 3–18.

——— AND ———, "Hours and Employment Variation in Business Cycle Theory," *Economic Theory* 1 (1991), 63–81.

MCGRATTAN, E. R., "A Progress Report on Business Cycle Models," *Quarterly Review* Federal Reserve Bank of Minneapolis 18 (1994a), 2–16.

———, "The Macroeconomic Effects of Distortionary Taxation," *Journal of Monetary Economics* 33 (1994b), 573–601.

MENDOZA, E. G., A. RAZIN, AND L. L. TESAR, "Effective Tax Rates in Macroeconomics: Cross-Country Estimates of Tax Rates on Factor Incomes and Consumption," *Journal of Monetary Economics* 34 (1994), 297–324.

ROTEMBERG, J. J. AND M. WOODFORD, "Oligopolistic Pricing and the Effects of Aggregate Demand on Economic Activity," *Journal of Political Economy* 100 (1992), 1153–1207.

WYNNE, M., "The Analysis of Fiscal Policy in Neoclassical Models," Research Paper 9212, Research Department, Federal Reserve Bank of Dallas, 1992.